# Paper Simulation II: shrinkage

N = 40 per arm; 30% expected MAR missingness; 40 imputations; 10000 replicates per scenario

| Scenario / metric | Naive | Single | Bonferroni<br>PS / PMM / RE | Holm<br>PS / PMM / RE | Proposed<br>PS / PMM / RE |
|---|---|---|---|---|---|
| Scenario I: joint FWER | 5.57 | 29.52 | 6.72/7.25/4.37 | 6.72/7.25/4.37 | 7.55/8.34/4.99 |
| Scenario II: T1 vs. T0 | 24.78 | 52.65 | 26.20/27.59/20.81 | 41.49/42.76/36.13 | 27.88/29.04/22.58 |
| Scenario II: T2 vs. T0 | 44.72 | 73.63 | 45.49/47.09/39.14 | 58.41/59.88/51.83 | 48.22/49.99/41.69 |
| Scenario II: T3 vs. T0 | 66.82 | 88.34 | 65.98/67.84/60.21 | 73.88/75.29/68.41 | 69.32/70.98/63.48 |
| Scenario II: T4 vs. T0 | 84.64 | 96.03 | 83.49/84.93/79.02 | 87.23/88.13/83.06 | 86.09/87.51/82.18 |
| Scenario II: T5 vs. T0 | 93.53 | 98.88 | 92.51/93.52/90.14 | 93.79/94.63/91.49 | 94.04/94.90/92.20 |
| Scenario II: joint power | 16.80 | 44.64 | 17.79/18.93/13.38 | 36.51/37.84/31.34 | 19.61/20.68/15.04 |

All entries are percentages. PS = propensity−score MI; PMM = predictive mean matching; RE = regression MI.

FWER = reject at least one null in Scenario I. Joint power = reject all five nulls in Scenario II.

Scenario I : 10000 successful, 0 failed | Scenario II : 10000 successful, 0 failed

# Paper Simulation II: shrinkage

N = 80 per arm; 30% expected MAR missingness; 40 imputations; 10000 replicates per scenario

| Scenario / metric | Naive | Single | Bonferroni<br>PS / PMM / RE | Holm<br>PS / PMM / RE | Proposed<br>PS / PMM / RE |
|---|---|---|---|---|---|
| Scenario I: joint FWER | 5.43 | 29.21 | 5.53/6.11/4.32 | 5.53/6.11/4.32 | 6.59/7.26/5.18 |
| Scenario II: T1 vs. T0 | 52.75 | 79.39 | 50.64/52.72/47.80 | 71.38/72.88/69.08 | 53.52/55.16/50.78 |
| Scenario II: T2 vs. T0 | 80.69 | 94.68 | 79.44/80.40/77.42 | 88.54/89.10/87.33 | 81.64/82.78/79.62 |
| Scenario II: T3 vs. T0 | 94.96 | 99.20 | 94.38/94.82/93.43 | 96.75/97.09/96.28 | 95.37/95.86/94.57 |
| Scenario II: T4 vs. T0 | 99.27 | 99.93 | 99.00/99.13/98.79 | 99.38/99.43/99.27 | 99.31/99.37/99.14 |
| Scenario II: T5 vs. T0 | 99.90 | 99.99 | 99.87/99.89/99.87 | 99.90/99.93/99.89 | 99.93/99.93/99.90 |
| Scenario II: joint power | 48.36 | 77.16 | 46.55/48.37/43.64 | 69.49/70.94/67.41 | 49.40/51.22/46.69 |

All entries are percentages. PS = propensity−score MI; PMM = predictive mean matching; RE = regression MI.

FWER = reject at least one null in Scenario I. Joint power = reject all five nulls in Scenario II.

Scenario I : 10000 successful, 0 failed | Scenario II : 10000 successful, 0 failed

# Paper Simulation II: non−shrinkage

N = 40 per arm; 30% expected MAR missingness; 40 imputations; 10000 replicates per scenario

| Scenario / metric | Naive | Single | Bonferroni<br>PS / PMM / RE | Holm<br>PS / PMM / RE | Proposed<br>PS / PMM / RE |
|---|---|---|---|---|---|
| Scenario I: joint FWER | 5.81 | 29.76 | 6.72/7.25/4.37 | 6.72/7.25/4.37 | 7.76/8.37/5.19 |
| Scenario II: T1 vs. T0 | 25.64 | 52.89 | 26.20/27.59/20.81 | 41.49/42.76/36.13 | 28.33/29.37/23.06 |
| Scenario II: T2 vs. T0 | 44.48 | 73.45 | 45.49/47.09/39.14 | 58.41/59.88/51.83 | 48.28/49.74/41.85 |
| Scenario II: T3 vs. T0 | 65.89 | 87.94 | 65.98/67.84/60.21 | 73.88/75.29/68.41 | 68.83/70.44/63.41 |
| Scenario II: T4 vs. T0 | 83.31 | 95.82 | 83.49/84.93/79.02 | 87.23/88.13/83.06 | 85.58/86.85/81.43 |
| Scenario II: T5 vs. T0 | 92.67 | 98.71 | 92.51/93.52/90.14 | 93.79/94.63/91.49 | 93.56/94.54/91.71 |
| Scenario II: joint power | 16.90 | 44.49 | 17.79/18.93/13.38 | 36.51/37.84/31.34 | 19.70/20.81/15.33 |

All entries are percentages. PS = propensity−score MI; PMM = predictive mean matching; RE = regression MI.

FWER = reject at least one null in Scenario I. Joint power = reject all five nulls in Scenario II.

Scenario I : 10000 successful, 0 failed | Scenario II : 10000 successful, 0 failed

# Paper Simulation II: non−shrinkage

N = 80 per arm; 30% expected MAR missingness; 40 imputations; 10000 replicates per scenario

| Scenario / metric | Naive | Single | Bonferroni<br>PS / PMM / RE | Holm<br>PS / PMM / RE | Proposed<br>PS / PMM / RE |
|---|---|---|---|---|---|
| Scenario I: joint FWER | 5.47 | 28.95 | 5.53/6.11/4.32 | 5.53/6.11/4.32 | 6.52/7.22/5.16 |
| Scenario II: T1 vs. T0 | 53.12 | 79.45 | 50.64/52.72/47.80 | 71.38/72.88/69.08 | 53.61/55.39/50.96 |
| Scenario II: T2 vs. T0 | 80.64 | 94.52 | 79.44/80.40/77.42 | 88.54/89.10/87.33 | 81.55/82.53/79.47 |
| Scenario II: T3 vs. T0 | 94.43 | 99.19 | 94.38/94.82/93.43 | 96.75/97.09/96.28 | 95.19/95.68/94.35 |
| Scenario II: T4 vs. T0 | 99.09 | 99.93 | 99.00/99.13/98.79 | 99.38/99.43/99.27 | 99.25/99.29/99.07 |
| Scenario II: T5 vs. T0 | 99.88 | 99.99 | 99.87/99.89/99.87 | 99.90/99.93/99.89 | 99.92/99.91/99.90 |
| Scenario II: joint power | 48.34 | 77.14 | 46.55/48.37/43.64 | 69.49/70.94/67.41 | 49.51/51.28/46.67 |

All entries are percentages. PS = propensity−score MI; PMM = predictive mean matching; RE = regression MI.

FWER = reject at least one null in Scenario I. Joint power = reject all five nulls in Scenario II.

Scenario I : 10000 successful, 0 failed | Scenario II : 10000 successful, 0 failed